Additional Implementation Details

A. Data Augmentation

Because of the limited number of labeled samples, data augmentation was applied to both downstream supervised training and self-supervised pretraining. The augmentation pipeline included random translation, random rotation, and Gaussian noise addition. These transformations were selected to preserve the essential spatial structure of the wave patterns while improving robustness to plausible acquisition variability. All augmentations were applied only to the training split.

Rotation angles were sampled uniformly from $(-30°, +30°)$, and translations were sampled uniformly as fractions of image width and height. The probabilities of applying rotation, translation, and noise addition were set to 0.6, 0.8, and 0.9, respectively. The noise percentage was sampled from the range $(0.3, 0.6)$. Noise was generated by smoothing white noise and scaling it by a randomly sampled standard deviation proportional to the value range of the corresponding channel. Noise was applied only within the ROI, and the resulting values were clamped to the valid range of each channel.

For the labeled downstream dataset, augmentation multiplicity depended on slice category: typical slices were augmented 12 times, whereas high-average slices were augmented 36 times. For the larger unlabeled pretraining dataset, each slice was randomly augmented four to six times with no distinction between slice categories because target viscoelastic maps were unavailable.

B. Region Normalized evaluation metric details

Let the target value range be divided into K distinct regions (bins). Let S_k be the set of pixel indices i where the ground truth value y_i falls into region k, and let N_k be the cardinality of S_k (the number of pixels in that region). The Region-Normalized metrics are defined as the arithmetic mean of the error calculated independently for each region (Eq. S1 and S2):

$$NMSE = \frac{1}{K} \sum_{k=1}^{K} \left(\frac{1}{N_k} \sum_{i \in S_k} (y_i - \hat{y}_i)^2 \right) \qquad \text{(S1)}$$

$$NMAE = \frac{1}{K} \sum_{k=1}^{K} \left(\frac{1}{N_k} \sum_{i \in S_k} |y_i - \hat{y}_i| \right) \qquad \text{(S2)}$$

Where y_i represents the ground truth value and $\hat{y}_i$ represents the predicted value at pixel i.

In this study, we set $K = 27$. The regions R_k were defined with a step size of 0.5 kPa. Specifically, the intervals were defined as $R_1 = (0, 0.5]$, increasing in 0.5 kPa increments up to the final region R_{27}, which encompassed all values $y_i > 13$ kPa. This discretization ensures that high complex shear modulus regions (high KPa), which characterize advanced fibrosis, contribute equally to the final score as low complex shear modulus regions (low KPa).